

New Development in Air Flow Measurements

Tracer gases have been used for many years for air flow testing in buildings. However, the expense and complexity of analysis equipment, and the difficulty of analyzing results, have restricted this kind of testing to research applications. For duct air flows, in particular, problems with getting uniform mixing and sampling also made tracer gas diagnostics unreliable. Recent research at Lawrence Berkeley National Laboratory (LBNL) has worked on addressing these problems—primarily to improve the measurement of air flows in commercial ducts. However, the techniques developed through this research can be applied just as usefully to measuring air handler flows in residential systems.

The LBNL study focused on using readily available inexpensive hardware to inject and sample tracer gas—usually CO₂—without the problems of poor mixing and sampling that have previously plagued this technique. With these problems resolved, and with the cost of high-accuracy CO₂ analyzers dropping, tracer gas techniques are quickly becoming both cost-effective and reliable outside of the research environment.

The tracer gas measurement technique follows traditional pulse injection and sample integration methods. A pulse of tracer is released over a few seconds and the downstream concentration is integrated over time. A simple mass balance equation is then used to determine the air flow. To ensure that the injected CO₂ mixes effectively in the airstream, LBNL developed whip injectors. These injectors are very small, flexible tubes that emit the CO₂ at high velocity by whipping around rapidly in the airstream, dispersing the tracer gas.

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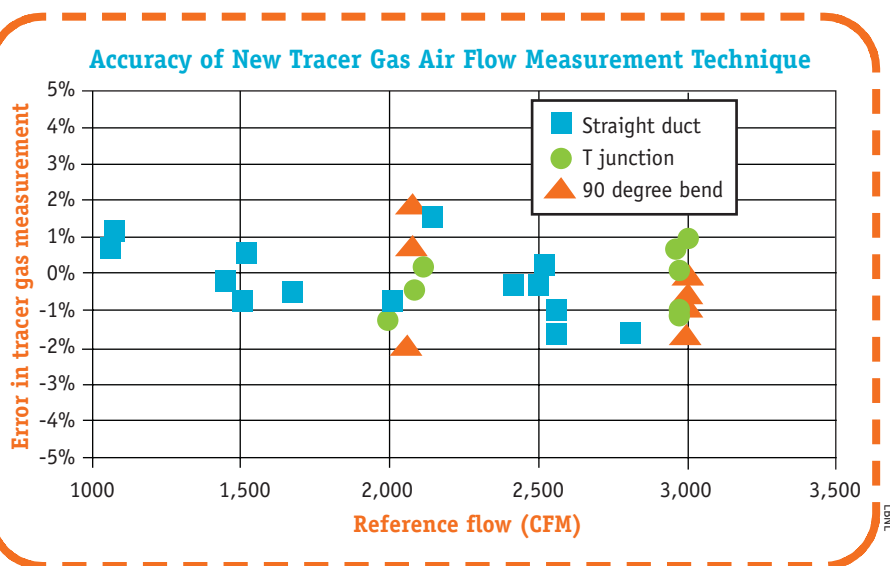


Figure 1. The results of 32 laboratory tests measuring air flows in straight rectangular ducts, with and without a T-junction upstream of the measurement section, indicate that the uncertainty of the test procedure using these injection and sampling techniques is less than 2%.



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result in a more uniform sampling of the air flow.

A data logger and computer analyze the sample; the only data needed for the analysis is the total mass of the injected pulse of tracer gas. Total measurement time, including analysis, is only a few minutes. Containers of known quantities of tracer gas can be prepared ahead of time to avoid the need for expensive items, such as a precision weighing scale, and the always awkward transportation of large high-pressure gas cylinders.

The results of 32 laboratory tests measuring air flows in straight rectangular ducts, with and without a T-junction upstream of the measurement section, indicate that the uncertainty of the test procedure using these injection and sampling techniques is less than 2% (see Figure 1). The reference flow meter has a rated accuracy of 0.5%.

This measurement procedure still needs to be tested in the residential sector to examine possible application issues, but the simplicity, relatively low cost, and insensitivity to upstream conditions mean that it has great potential.

—**Iain Walker**

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For more information:

To learn more about the equipment and techniques described above, contact Craig Wray, mechanical engineer at LBNL, at cpwray@lbl.gov.